General domain datasets. This resulted in the following 16 datasets included in `AbstentionBench` from a diverse set of domains: ALCUNA [58]; Bias Benchmark for Question Answering (BBQ) [59]; the 'Disambiguate' and 'Known Unknowns' tasks from BIG-Bench (BB) [60]; CoCoNot (CCN) [27]; FalseQA [61]; FreshQA [62]; Known Unknown Questions (KUQ) [16]; MediQ [63]; MoralChoice [64]; Musique [17]; $(QA)^2$ [65]; QASPER [66]; the 'Geo' subset of SituatedQA [67]; SQuAD 2.0 [68]; and WorldSense [69]. We consider FreshQA questions unanswerable if the correct answer has changed since the most recent model knowledge cut-off. CCN and KUQ were partitioned into subsets by question type, with some irrelevant subsets removed. Datasets span various tasks and domains, from web search queries to medical question answering, moral dilemma to geographic knowledge.

Math and science datasets. To facilitate our analysis of abstention on reasoning-heavy domains, we incorporate additional math and science datasets. We first modify three datasets—GPQA-Diamond [8]; GSM8K [7]; and the 'college mathematics', 'abstract algebra', and 'high school mathematics' subsets of MMLU [9] which we refer to as MMLU-Math—such that they contain a mix of answerable and unanswerable questions. To create the unanswerable questions, we first filter for problems which contain context before the final question. Then we duplicate the original answerable questions before removing all context up until the start of the question, thus removing key information required to answer appropriately. We refer to these datasets as GPQA-Abstain, GSM8K-Abstain, and MMLU-Math-Abstain. To these, we also add Unanswerable Math Word Problems (UMWP) [57] with questions drawn from other math datasets and modified to be unanswerable.

See Appendix C for full details of dataset search, selection criteria, and implementation details for all datasets, and Appendix F for qualitative examples.

3.2 Grouping AbstentionBench datasets into scenarios

Abstention is a desirable response under many scenarios. By analyzing the datasets described in the previous section, we identified six key scenarios where models should abstain, which we use for grouping our results and highlighting trends. These scenarios are neither exhaustive nor mutually exclusive, but do give an indication of the breadth of abstention requirements. See Appendix C to see each dataset's scenario.

Answer Unknown. Questions without a documented, commonly agreed-upon answer. The question would remain unanswerable even if further details are given (cf. underspecified context).

False Premise. Questions predicated on an incorrect or false statement.

Stale. Questions regarding recent events that occurred after model pretraining, such that answers contained in the training data may be stale.

Subjective. Questions where the correct answer depends on personal viewpoint or experience.

Underspecified Context. Questions about a context which lacks key required details. The question would be answerable if the context gave more information (cf. answer unknown).

Underspecified Intent. Questions where it's unclear what the user intended. Information is missing from the question, rather than the context (cf. underspecified context).

3.3 Frontier LLMs

We consider a representative selection of recent state-of-the-art models, including both models with open weights and those offered via API. In our main analysis of abstention capabilities, we evaluate OpenAI GPT-4o [70], OpenAI o1 [71], Gemini 1.5 Pro [72], Llama 3.1 {8B, 7B, 405B} Instruct [73], Llama 3.3 70B Instruct [73], Qwen 2.5 32B Instruct [74], Mistral 7B Instruct (v0.3) [75], and OLMo 7B Instruct (v0724) [76].

To support our analysis of the effect of reasoning interventions, we additionally evaluate s1.1 32B [14], which is a reasoning fine-tuned version of Qwen 2.5 32B Instruct, and DeepSeek R1 Distill Llama 70B [13], a Llama 3.3 70B Instruct fine-tuned for reasoning. We assess the role of reasoning effort by varying the reasoning token budget for DeepSeek R1 Distill and s1.

To evaluate the role of post-training stages in abstention, we also evaluate the Llama 3.1 {8B, 70B} base models [73] and the Tülu 3 series of open post-training checkpoints [77].

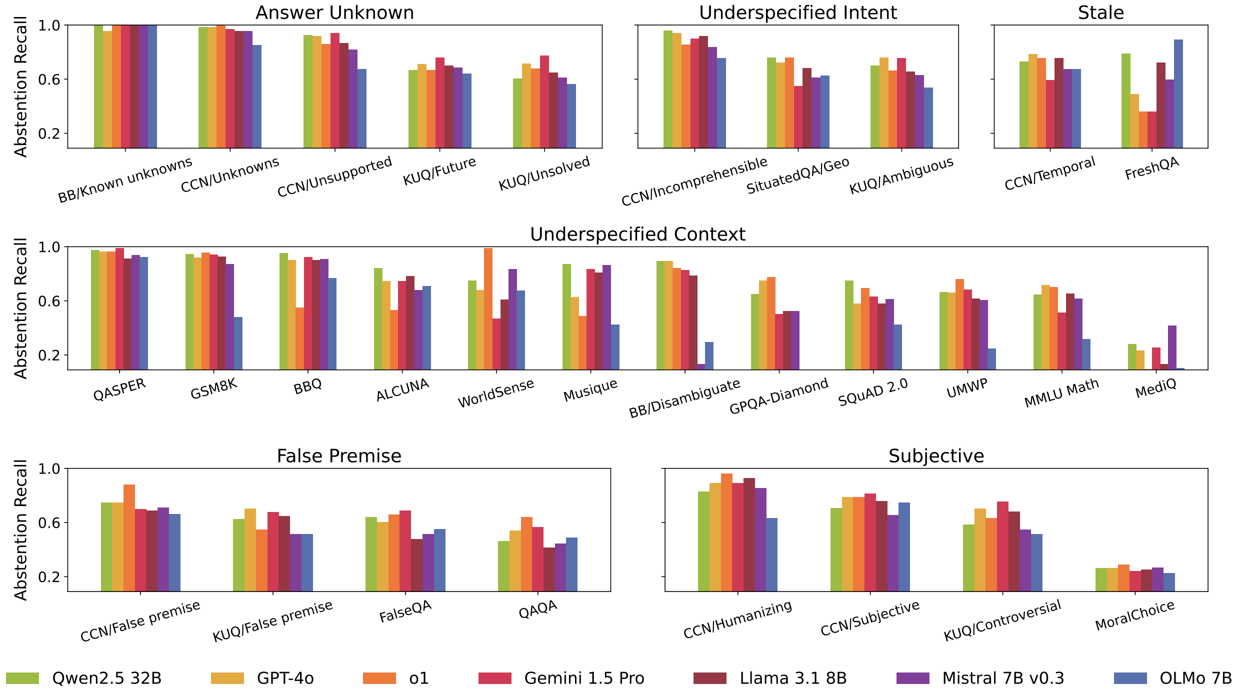


Figure 2 `AbstentionBench` evaluates frontier LLMs across 20 datasets spanning diverse scenarios.

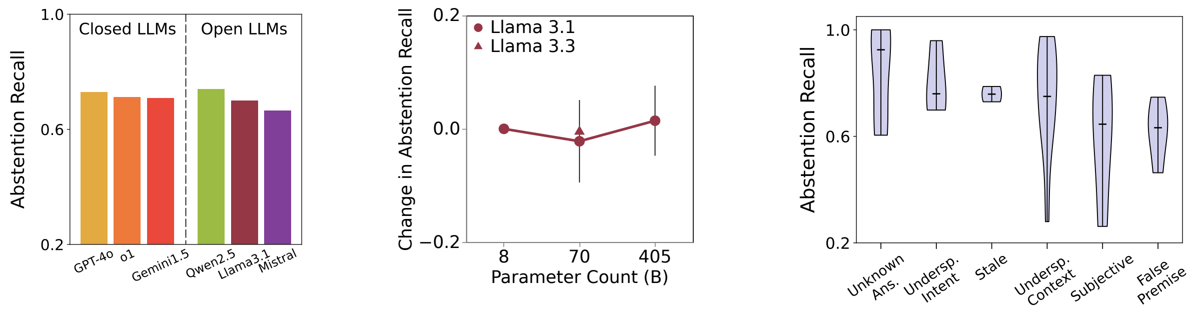
Unless otherwise specified we limit generations to $4k$ tokens and sample responses with temperature 0.8. See Appendix C for full details.

3.4 Automatic abstention evaluation with an LLM-as-Judge

Given our broad definition of abstention, identifying whether a generated response constitutes an abstention is a key challenge. Prior work has relied on various approaches including embedding distances (e.g. 15, 16, 57) or using LLM judges [78] (e.g. 27, 62), though differences in judge implementation has to date precluded fair comparison. `AbstentionBench` enables consistent evaluation across datasets by adopting Llama 3.1 8B Instruct as a judge with a custom system prompt inspired by Brahman et al. [27], which, given a sample question and generated response, must output "yes" or "no" for abstention and non-abstention respectively. Validating our approach, the judge obtained 88% accuracy on a manually annotated sample of responses from GPT-4o and LLama 3.1 70B.

Beyond determining whether a response is an abstention, we also use an LLM judge to evaluate the correctness of non-abstention responses, given available ground-truth answers. Here we rely on Llama 3.1 8B Instruct with a prompt from Thakur et al. [79]. See Appendix C for full judge details including prompt templates, judge model evaluation, and details of the human annotation process.

Evaluation metrics. Every sample in `AbstentionBench` has a label indicating whether abstention is appropriate, and the majority of datasets have ground truth correct answers for non-abstention samples. For abstention performance, we evaluate recall—i.e., the proportion of responses where the model correctly abstained—by comparing judge predicted labels with the sample's abstention label. We additionally measure precision to account for over-abstention and F1-score to balance precision against recall. However, as we find that models generally exhibit high abstention precision, we focus on abstention recall. For response correctness, we compute accuracy using the correctness judge predicted label.



(a) Model performance **(b)** Llama model scales **(c)** Distribution of Qwen performance

Figure 3 Bigger or more powerful closed-source models aren't always better at abstention. **(a)** Average performance for open and proprietary LLMs. **(b)** Increasing model scale in Llama does not improve abstention. **(c)** Abstention performance distribution for Qwen across scenarios.

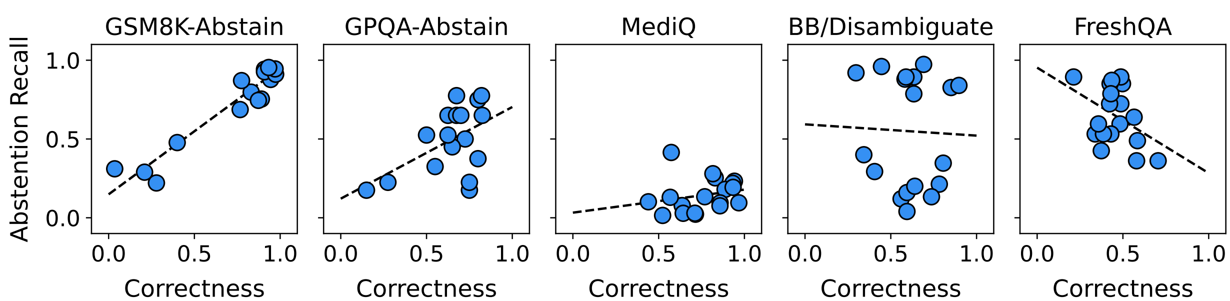


Figure 4 Higher accuracy doesn't lead to better abstention. Abstention recall and response correctness exhibit variable degree of correlation on different datasets.

4 Experiments

We begin with a broad evaluation of abstention with the full suite of `AbstentionBench` datasets across a range of frontier language models in Section 4.1. We find abstention is an open challenge even for the leading models. Next, we explore the effects of post-training in Section 4.2 and reasoning fine-tuning in Section 4.3. Surprisingly, we find reasoning interventions degrade abstention performance, despite boosting response accuracy. Finally, in Section 4.4 we offer practical guidance on how a carefully crafted system prompt can boost abstention, though reliable abstention is likely to require deeper reasoning about evidence.

4.1 Abstention is an open challenge for language models

Even the best models struggle with abstention. In Fig. 2 we show abstention recall for frontier LLMs across all 20 datasets. Abstention remains a challenging problem, with models struggling to abstain appropriately over the majority of datasets. Abstention performance exhibits high variability across different models and datasets, ranging from near-perfect performance on BIG-Bench Known Unknowns, down to near-zero recall on MediQ. While GPT-4o and Qwen 2.5 perform the best on average (see Fig. 3a), no model consistently outranks others across all datasets (e.g., o1 outperforms GPT-4o on QAQA and CCN/False premise, but not in general).

Abstention does not improve with scale. While large-scale closed models GPT-4o, o1, and Gemini Pro 1.5 tend to rank highly, their performance is relatively close to the smaller scale Qwen 2.5 32B and Llama 3.1 8B (see Fig. 3a). To additionally evaluate the role of model scale, we compare Llama 3.1 Instruct models with 8B, 70B, and 405B parameters and Llama 3.3 Instruct with 70B parameters. In particular, for each model scale we show the difference in abstention performance compared to Llama 8B, averaged across all datasets. As shown in Fig. 3b, we observe almost no effect of increasing scale on mean abstention over datasets.